

Anubhav Gupta

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EDUCATION

- **University of Maryland, College Park**
PhD, Computer Science College Park, MD
- **University of Maryland, College Park**
MS, Computer Science College Park, MD
- **Indian Institute of Technology, Delhi**
B.Tech., Electrical Engineering New Delhi, India

EXPERIENCE

- **University of Maryland, College Park** Jan'21 - Present
Graduate Assistant College Park, MD
 - Advisor: [Prof. Abhinav Shrivastava](#)
 - Current Research Interests: Video and action understanding and transferring the actions to robots
- **Amazon** May'24 - Aug'24
Applied Scientist Intern Sunnyvale, CA
 - Diffusion-based multi-view image editing for indoor scenes
 - Conceptualized and created a large-scale dataset starting from 3D-Front
- **Amazon** May'23 - Aug'23
Applied Scientist Intern Sunnyvale, CA
 - Diffusion based key-stroke assisted human body editing (pose and body shape transformations). The intended application was a 2D virtual try-on
 - Worked with Stable Diffusion 1.5, and used ControlNets to modulate the latent space
- **Amazon** May'22 - Aug'22
Applied Scientist Intern Seattle, WA
 - Worked on edge-based computer vision product focusing on objection detection
 - Built a prototype model and pushed it to production for field trial
 - Received a return offer for a full-time position
- **Swiggy** |  Sep'20 - Jan'21
Data Scientist Bangalore, India
 - Deep learning-based road network extraction from satellite imagery
 - Extracted building footprints at scale from OSM and constructed Point of Interest Polygons
- **Netradyne Technologies** |  Jun'17 - Aug'20
Senior Research Engineer Bangalore, India
 - Object Detection: Curated the datasets and trained the object detection models for different geographies.
 - Optimization and Engineering: model acceleration and compression for on-device analytics
 - Infrastructure Development: Primary developer for analytics video data lake and model evaluation tool

PUBLICATIONS

C=CONFERENCE, J=JOURNAL, S=IN SUBMISSION, T=THESIS

- [C.1] **Anubhav Gupta**, et al. (2026). **VidParse: Online Parsing of Egocentric Procedures Like a Pro** . In *ECCV 2026*
We propose VidParse, an online, training-free framework that parses egocentric video into procedural action steps by combining manipulation-anchored features from frozen foundation models with task-graph constrained beam search.
- [C.2] Shuaiyi Huang, Mara Levy **Anubhav Gupta**, et al. (2025). **TREND: Tri-teaching for Robust Preference-based Reinforcement Learning with Demonstrations** . In *ICRA 2025*
Preference feedback collected by human or VLM annotators is often noisy, presenting a significant challenge for preference-based reinforcement learning. To address this challenge, we propose TREND, a novel framework that integrates few-shot expert demonstrations with a tri-teaching strategy for effective noise mitigation
- [C.3] Gowthami Somepalli*, **Anubhav Gupta***, et al. (2024). **Measuring Style Similarity in Diffusion Models** . In *ECCV 2024*
Can we measure the style-similarity between images? We propose Contrastive Style Descriptors (CSD) as a method to represent the style of an image. Using this model, we study the style replication in image generation models.
- [C.4] Shihira R Maiya*, **Anubhav Gupta***, et al. (2024). **Latent-INR: A Flexible Framework for Implicit Representations of Videos with Discriminative Semantics** . In *ECCV 2024*
We show semantic capabilities in Implicit Neural Representations (INR) by proposing a novel framework that learns discriminative semantics in videos.
- [C.5] Archana Swaminathan, **Anubhav Gupta**, et al. (2024). **LEIA: Latent View-invariant Embeddings for Implicit 3D Articulation** . In *ECCV 2024*
Modeling unseen 3D articulation states by interpolating across a learnable, view-invariant latent space.
- [C.6] Kamal Gupta, Gowthami Somepalli, **Anubhav Gupta**, et al. (2021). **PatchGame: Learning to Signal Mid-level Patches in Referential Games** . In *NeurIPS 2021*
Emergent communication via mid-level patches in a referential game played on a large-scale image dataset
- [C.7] Abhinav Ganesan, **Anubhav Gupta**, Jose Mathew et al. (2021). **Mining Points of Interest via Address Embeddings: An Unsupervised Approach** . In *LocalRec 2021*
Unsupervised PoI mapping (polygon boundaries) using GPS, OpenStreetMaps and Address Information in highly dense environments

PATENTS

- [1] Michael Campos, **Anubhav Gupta**, et al. (2019). **Detection of driving actions that mitigate risk** . US, Patent No. 10782654 Publication Date: 2020/9/22

SERVICE

- Reviewer, ICML 2026
- **Outstanding Reviewer**, CVPR 2025
- Reviewer, ICRA 2025
- Reviewer, CVPR 2024
- Workshop Organizing Committee Member: Open World Vision, CVPR 2023, Vancouver
- **Workshop Co-organizer**: Dealing With Novelty in the Open Worlds, WACV 2023, Hawaii
- Workshop Organizing Committee Member: Open World Vision, CVPR 2022, New Orleans
- **Workshop Co-organizer**: Dealing With Novelty in the Open Worlds, WACV 2022, Hawaii